

TasteRank Explorer

Data Appendix

Source Data, Tables, and Reference Materials

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Companion documents: Summary · Technical Appendix · Methods Primer · Grape Reference

D.1 Sensory Dimension Key

The following abbreviations are used in the data tables. All scores are integers on a 0–5 scale representing canonical varietal typicality across 13 sensory dimensions drawn from established wine-tasting methodology (UC Davis, Peynaud, and related structured tasting frameworks).

Abbr.	Dimension	Description
Col	Colour Depth	Intensity of pigmentation (pale to opaque)
Aro	Aromatic Intensity	Strength of nose from low to pronounced
Flo	Floral	Degree of floral aromatic contribution
Frt	Fruit Ripeness	Ripe/dried fruit vs. underripe/green fruit
Hrb	Herbal/Earthy	Green herb, earth, mineral notes
Spk	Spice/Oak	Oak-derived and spice character
Acd	Acidity	Perceived acidity from low to high
Tan	Tannin	Tannin level (reds); 0 for most whites
Bod	Body	Weight and texture on palate
Alc	Alcohol	Perceived alcohol warmth
Fla	Flavour Intensity	Palate flavour concentration
Fin	Finish	Length and persistence of aftertaste
Cpx	Complexity	Layeredness and evolution of character

D.2 Community Summary

Six communities were detected by the Clauset–Newman–Moore greedy modularity optimisation algorithm. The table below summarises each community’s character, size, colour composition, and hub varieties.

ID	Character	Size	R / W	Hub Varieties
C0	Full-bodied Mediterranean Reds	31	31 / 0	Sagrantino, Nero d’Avola, Lagrein, Negroamaro, Plavac Mali
C1	Light / Aromatic Cross-boundary	21	4 / 17	Assyrtiko, Falanghina, Albariño, Riesling, Kadarka
C2	Mid-weight Structured Reds	18	18 / 0	Limniona, St. Laurent, Zweigelt, Blaufränkisch, Sangiovese
C3	Rich Textural Whites	13	0 / 13	Marsanne, Fiano, Godello, Pinot Gris, Chardonnay
C4	Mineral / Crisp Whites	10	0 / 10	Greco, Friulano, Verdicchio, Verdejo, Vermentino
C5	Lean Neutral & Aromatic Whites	8	0 / 8	Gewürztraminer, Petit Manseng, Muscat Blanc, Malvasia

D.3 Pipeline Parameters

A consolidated reference of all parameters used in the analytical pipeline.

Parameter	Value
Number of varieties n	101 (53 red, 48 white)
Sensory dimensions	13 (scored 0–5 integers)
Similarity metric	Cosine on raw sensory vectors
k -NN graph k	5
Symmetrisation	Maximum-weight union
Eigenvector centrality solver	Power iteration via NetworkX
max_iter	2000
tol	10^{-6}
PageRank damping α	0.85
Community detection algorithm	Clauset–Newman–Moore greedy modularity
Number of communities detected	6
Final modularity Q	≈ 0.41
Spectral gap $ \lambda_2/\lambda_1 $	≈ 0.72

D.4 Source Data Files

The TasteRank Explorer is shipped as a single self-contained HTML file. All variety profiles, pre-computed centrality scores, community assignments, and edge data are embedded directly in the JavaScript source of the page; no external data files are required at runtime.

For programmatic re-analysis, the canonical sensory profile data is reproduced in Section D.5 of this appendix and can be parsed directly from the table.

D.5 Reproducibility

The analytical pipeline is deterministic given the sensory profile data and the random seed. Software versions: Python 3.11, NetworkX 3.2, NumPy 1.26. Convergence of power iteration to tolerance 10^{-6} is achieved in 35–45 iterations from a uniform initial vector. The Clauset–Newman–Moore algorithm produces the same six-community partition across replicate runs.

D.6 Source Materials

The 13 sensory dimensions and the 0–5 scoring scale draw on the structured tasting tradition that runs from Peynaud through the modern descriptive frameworks (UC Davis, AWRI, the Wine & Spirit Education Trust). Variety scores are calibrated against the standard ampelographic literature, principally Robinson, Harding & Vouillamoz (*Wine Grapes*), d’Agata (*Native Wine Grapes of Italy*), and Galet (*Dictionnaire encyclopédique des cépages*). The companion document *Grape Reference* provides per-variety descriptive notes.

D.7 Complete Variety Data (Ranks 1–101)

All 101 grape varieties sorted by TasteRank. Columns show rank, variety name, type (R/W), community assignment, TasteRank score, and the 13 sensory profile dimensions (see D.1 for abbreviations).

#	Variety	T	Com	TR	Col	Aro	Flo	Frnt	Hrb	Spk	Acid	Tan	Bod	Alc	Fla	Fin	Cpx
1	Sagrantino	R	C0	0.3020	5	3	1	4	2	3	3	5	5	4	4	4	4
2	Nero d’Avola	R	C0	0.2849	5	3	1	4	2	2	3	4	4	4	4	3	3
3	Lagrein	R	C0	0.2748	5	3	1	3	2	2	3	4	4	3	3	3	3
4	Negroamaro	R	C0	0.2380	4	3	1	4	2	2	3	3	4	4	3	3	3
5	Plavac Mali	R	C0	0.2251	5	3	1	4	2	2	3	4	5	5	4	3	3
6	Montepulciano	R	C0	0.2108	5	3	1	4	2	2	3	4	4	3	3	3	3
7	Petit Verdot	R	C0	0.2093	5	3	2	3	2	3	3	5	5	4	4	4	4
8	Monastrell	R	C0	0.2060	5	4	1	4	2	3	2	4	5	5	4	4	3
9	Petite Sirah	R	C0	0.1768	5	3	1	4	1	3	3	5	5	4	4	4	3
10	Pinotage	R	C0	0.1710	5	4	0	4	1	3	3	4	4	4	4	3	3
11	Malbec	R	C0	0.1697	5	3	2	4	1	3	3	4	5	4	4	4	3
12	Tannat	R	C0	0.1676	5	3	1	3	2	3	4	5	5	4	4	4	4
13	Garnacha Tintorera	R	C0	0.1670	5	3	1	4	2	2	3	4	5	4	3	3	2
14	Dolcetto	R	C0	0.1659	4	3	1	3	1	1	3	3	3	3	3	2	2
15	Mourvèdre	R	C0	0.1565	5	4	1	3	3	3	3	4	5	4	4	4	4
16	Primitivo	R	C0	0.1498	5	4	1	5	1	3	2	3	5	5	4	3	3
17	Zinfandel	R	C0	0.1498	5	4	1	5	1	3	2	3	5	5	4	3	3
18	Merlot	R	C0	0.1333	4	3	1	4	2	3	3	3	4	4	3	4	4
19	Carignan	R	C0	0.1283	4	3	1	3	3	1	4	4	4	3	3	3	3
20	Limniona	R	C2	0.1223	4	4	2	3	3	2	4	4	4	3	4	4	4
21	St. Laurent	R	C2	0.1218	3	3	1	3	2	2	4	3	3	3	3	3	3
22	Bobal	R	C0	0.1212	5	3	1	3	2	1	4	4	4	3	3	3	3
23	Grenache	R	C0	0.1206	3	3	1	4	2	2	2	2	4	5	3	3	3
24	Syrah	R	C0	0.1177	5	4	2	3	2	4	3	4	5	4	4	5	5
25	Tempranillo	R	C0	0.1163	4	3	1	3	3	3	3	4	4	3	3	4	4
26	Mavrud	R	C0	0.1149	5	3	1	3	3	2	4	5	4	4	4	4	4
27	Zweigelt	R	C2	0.1134	3	3	1	3	1	1	3	2	3	3	3	3	2
28	Counoise	R	C2	0.1121	3	3	2	3	2	1	4	3	3	3	3	3	3
29	Graciano	R	C0	0.1027	5	4	2	3	3	3	4	5	4	3	4	4	4
30	Marsanne	W	C3	0.0978	3	3	1	3	1	2	2	0	4	4	3	3	3
31	Cabernet Sauvignon	R	C0	0.0940	5	4	1	3	3	4	3	5	5	4	4	5	5
32	Tinta Roriz	R	C0	0.0925	4	3	1	3	3	3	3	4	4	3	3	4	4
33	Blafränkisch	R	C2	0.0909	4	4	1	3	2	2	4	4	3	3	4	4	4
34	Fiano	W	C3	0.0855	2	4	2	3	1	1	4	0	3	3	4	4	4
35	Touriga Nacional	R	C0	0.0849	5	5	3	3	2	3	3	5	5	4	5	5	5
36	Carménère	R	C2	0.0839	4	4	1	3	4	2	3	3	4	3	4	3	3
37	Godello	W	C3	0.0826	2	3	1	3	1	1	4	0	3	3	3	3	4
38	Pinot Gris	W	C3	0.0803	3	3	1	3	0	1	3	0	3	3	3	3	3

#	Variety	T	Com	TR	Col	Aro	Flo	Fr	Hrb	Spk	Ac	Tan	Bod	Alc	Fla	Fin	Cpx
39	Aglianico	R	C0	0.0787	4	4	1	3	3	3	5	5	4	4	4	5	5
40	Baga	R	C0	0.0787	4	3	1	2	3	2	5	5	3	3	3	4	4
41	Chardonnay	W	C3	0.0764	3	3	1	3	1	3	3	0	4	4	3	4	4
42	Sémillon	W	C3	0.0753	3	3	1	3	0	2	2	0	4	4	3	4	4
43	Encruzado	W	C3	0.0747	2	3	1	3	1	2	4	0	3	3	3	4	4
44	Greco	W	C4	0.0742	2	3	1	2	1	0	4	0	3	3	3	3	4
45	Roussanne	W	C3	0.0710	3	4	2	3	2	2	3	0	4	3	4	4	4
46	Friulano	W	C4	0.0688	2	3	1	2	1	0	3	0	3	3	3	3	3
47	Verdelho	W	C3	0.0683	2	3	1	3	0	1	3	0	3	3	3	3	3
48	Verdicchio	W	C4	0.0674	2	3	1	2	1	0	4	0	3	3	3	3	3
49	Cabernet Franc	R	C0	0.0671	3	4	2	2	4	2	4	4	3	3	4	4	4
50	Verdejo	W	C4	0.0662	2	3	1	2	2	0	4	0	3	3	3	3	3
51	Barbera	R	C2	0.0650	4	3	1	3	1	2	5	2	3	3	3	3	3
52	Vermantino	W	C4	0.0650	2	3	1	2	2	0	4	0	3	3	3	3	3
53	Corvina	R	C2	0.0641	3	3	2	3	1	1	4	2	3	3	3	3	3
54	Sangiovese	R	C2	0.0632	3	4	1	2	3	2	5	4	3	3	4	4	4
55	Mencia	R	C2	0.0601	3	4	2	2	3	2	4	3	3	3	4	3	4
56	Grüner Veltliner	W	C4	0.0594	2	3	1	2	2	0	4	0	3	3	3	3	4
57	Pinot Blanc	W	C3	0.0574	2	2	1	2	0	1	3	0	3	3	2	3	2
58	Nebbiolo	R	C2	0.0567	3	5	3	3	3	3	5	5	3	4	5	5	5
59	Arneis	W	C3	0.0560	2	3	2	3	0	0	3	0	3	3	3	3	3
60	Vidiano	W	C3	0.0550	2	3	2	3	1	1	3	0	3	3	3	3	3
61	Cinsault	R	C2	0.0540	2	3	2	3	1	1	3	2	2	3	3	2	2
62	Touriga Franca	R	C2	0.0524	4	4	3	3	1	2	3	3	4	3	4	4	4
63	Falanghina	W	C1	0.0515	2	3	2	2	0	0	4	0	3	3	3	3	3
64	Nerello Mascalese	R	C2	0.0499	2	4	2	2	3	2	5	3	2	3	4	4	5
65	Garganega	W	C4	0.0497	2	2	1	2	1	0	3	0	3	3	2	3	3
66	Silvaner	W	C4	0.0487	2	2	1	2	1	0	3	0	3	3	2	3	3
67	Xinomavro	R	C2	0.0481	3	4	1	2	3	2	5	5	3	3	4	5	5
68	Clairette	W	C4	0.0437	2	2	1	2	1	0	3	0	3	3	2	2	2
69	Assyrtiko	W	C1	0.0409	2	4	1	2	1	0	5	0	3	3	4	4	5
70	Viognier	W	C3	0.0400	3	5	4	4	0	2	2	0	4	4	4	3	4
71	Pinot Noir	R	C2	0.0399	2	4	2	2	2	2	4	2	2	3	4	4	5
72	Chenin Blanc	W	C4	0.0391	2	3	2	3	1	1	5	0	3	3	3	4	4
73	Albariño	W	C1	0.0387	1	4	2	2	1	0	5	0	2	3	3	3	4
74	Gewürztraminer	W	C5	0.0361	3	5	4	4	1	2	2	0	4	4	5	3	4
75	Furmint	W	C1	0.0361	2	3	1	2	1	1	5	0	3	3	3	4	5
76	Petit Manseng	W	C5	0.0356	2	4	2	4	1	1	5	0	3	3	4	4	4
77	Riesling	W	C1	0.0353	1	4	3	2	0	0	5	0	2	2	4	4	5
78	Trousseau	R	C2	0.0352	2	3	1	2	2	1	4	2	2	2	3	3	3
79	Kadarka	R	C1	0.0335	2	3	2	2	2	1	4	2	2	2	3	2	3
80	Loureiro	W	C1	0.0324	1	4	3	2	0	0	4	0	2	2	3	2	3
81	Ribolla Gialla	W	C1	0.0318	1	3	1	2	1	0	4	0	2	2	3	2	3
82	Malvasia	W	C5	0.0315	2	4	3	3	1	0	3	0	3	3	3	3	3
83	Gamay	R	C2	0.0310	2	4	2	2	1	1	4	1	2	2	3	2	3

#	Variety	T	Com	TR	Col	Aro	Flo	Frt	Hrb	Spk	Acid	Tan	Bod	Alc	Fla	Fin	Cpx
84	Muscat Blanc	W	C5	0.0302	2	5	5	3	0	0	3	0	2	2	4	2	3
85	Malagousia	W	C5	0.0297	2	4	4	3	1	0	3	0	3	3	4	3	4
86	Torrontés	W	C5	0.0296	2	5	4	3	0	0	3	0	3	3	4	2	3
87	Sauvignon Blanc	W	C1	0.0290	1	4	1	2	4	0	5	0	2	2	4	3	3
88	Cortese	W	C1	0.0256	1	2	1	2	0	0	4	0	2	2	2	2	2
89	Arinto	W	C1	0.0255	1	3	1	2	0	0	5	0	2	2	3	3	3
90	Welschriesling	W	C1	0.0234	1	3	2	2	0	0	4	0	2	2	2	2	2
91	Müller-Thurgau	W	C1	0.0226	1	3	2	2	0	0	3	0	2	2	2	2	2
92	Schiava	R	C1	0.0219	1	3	2	2	1	0	4	1	1	2	2	2	2
93	Poulsard	R	C1	0.0219	1	3	2	2	1	0	4	1	1	2	2	2	2
94	Colombard	W	C1	0.0214	1	2	0	2	1	0	4	0	2	2	2	2	2
95	Chasselas	W	C5	0.0209	1	2	0	2	0	0	3	0	2	2	2	2	2
96	Trebbiano	W	C5	0.0196	1	2	0	2	0	0	3	0	2	2	2	2	2
97	Frappato	R	C1	0.0195	2	4	3	2	1	0	4	1	2	2	3	2	3
98	Aligoté	W	C1	0.0188	1	2	0	2	0	0	4	0	2	2	2	2	2
99	Melon de Bourgogne	W	C1	0.0174	1	2	0	1	0	0	4	0	2	2	2	2	2
100	Mauzac	W	C1	0.0167	1	2	1	2	0	0	4	0	2	2	2	2	2
101	Ugni Blanc	W	C5	0.0101	1	1	0	1	0	0	5	0	1	2	1	1	1

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